SYSTEM UNIT COMPONENTS

Complete Exam-Oriented Study Notes

2.1 The Motherboard | 2.2 Processor | 2.3 Memory

Easy wording • Detailed explanations • Diagrams • Tables • Revision questions

How to Use These Notes

These notes are written for exam preparation. Learn the definition first, understand the function, and then practise explaining the concept in your own words. For a long question, use a short introduction, labelled diagram where relevant, headings for the main points, and a concluding sentence.

- For a 2–3 mark question: give a precise definition and two relevant points.
- For a 5 mark question: define the term, explain its main parts or functions, and include an example.
- For a long question: organise the answer under headings and draw the relevant diagram neatly.
- Do not confuse main memory (RAM) with storage (SSD/HDD). They serve different purposes.

Contents

• 2.1 The Motherboard

• 2.2 Processor (CPU): Control Unit, ALU, Machine Cycle, System Clock, Comparison of PC Processors

• 2.3 Memory: Bits, Bytes, Addressing, RAM, ROM, Cache, Flash and CMOS

• Quick Comparison Tables

• Exam-Focused Short Questions and Answers

• Long Questions / Practice Paper

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2.1 The Motherboard

Definition

The motherboard is the main printed circuit board (PCB) inside a computer. It provides the physical connections and communication pathways that allow the processor, memory, storage devices, expansion cards and other components to work together.

Easy way to remember: The motherboard is like the main road network of a city. Components are connected to it, and data travels between them through electronic pathways called buses and links.

Main Functions of the Motherboard

1. Holds and connects major components such as the CPU, RAM and expansion cards.
2. Provides pathways for data, address and control signals to move between components.
3. Distributes electrical power from the power supply to components through appropriate connectors and voltage-regulation circuits.
4. Provides connectors for storage devices, USB ports, audio, networking and display-related interfaces.
5. Contains firmware storage used to start the computer and initialise hardware.
6. Supports expansion and upgrades, subject to compatibility.

Important Motherboard Components

Component	Purpose	Exam point
CPU socket	Holds the processor and provides its electrical connection.	Socket type must match the CPU.
RAM slots (DIMM slots)	Hold memory modules used by running programs.	Motherboard supports specific memory types and capacities.
Chipset / platform controller	Coordinates selected input/output functions and connectivity.	Exact duties differ by platform and generation.
Expansion slots (PCI Express)	Accept graphics, network, sound or other add-in cards.	PCIe slot size and available lanes matter.
Storage connectors	Connect SATA drives or M.2 devices.	M.2 is a form factor; it may use SATA or PCIe/NVMe.
Power connectors	Bring power from the power supply to the board and CPU.	Never force a connector in the wrong orientation.
Firmware chip	Stores UEFI/BIOS firmware used during startup.	Firmware initialises hardware and begins booting.
Rear I/O ports and internal headers	Connect USB, audio, network, front-panel buttons and other devices.	Port availability varies by model.
CMOS/RTC battery	Helps maintain real-time clock and selected firmware settings when power is removed.	Many settings are stored in non-volatile firmware memory; the battery mainly supports the clock and legacy CMOS-backed settings.

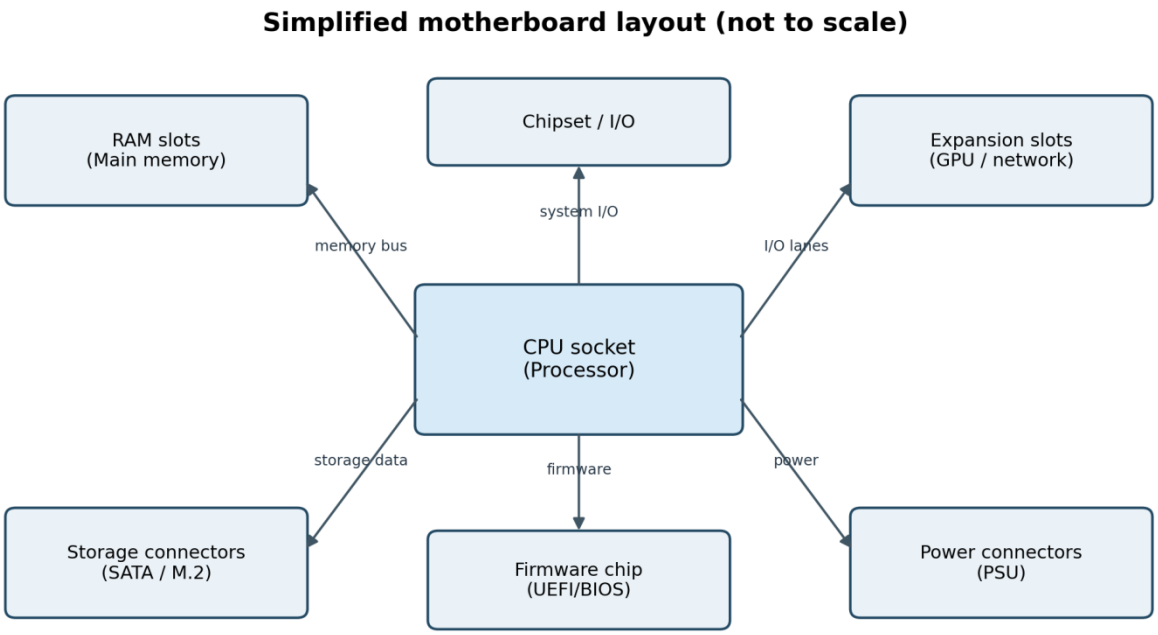


Figure 1. Simplified motherboard layout. Real boards vary by model; this diagram is conceptual.

Buses and Communication

A bus is a communication pathway used to transfer information between computer components. Modern systems use a mixture of shared buses and point-to-point links, but the traditional bus model is useful for understanding the basic idea.

Bus / signal type	What it carries	Simple example
Data bus	The actual data or instruction bits being transferred.	A value being read from RAM.
Address bus	The address that identifies the memory location or device being accessed.	The CPU specifies which memory location to read.
Control signals	Commands and timing/status signals that coordinate an operation.	Read, write, interrupt or ready signals.

Important distinction: The address tells the system where to access; the data is what is transferred; control signals tell the components what operation to perform.

Motherboard Form Factors

A form factor defines the board's physical size, mounting points and often its general connector layout. Common examples include ATX, Micro-ATX and Mini-ITX.

Form factor	General characteristics	Typical use
ATX	Larger board with room for more expansion and connectors.	Desktop towers and workstations.
Micro-ATX	Smaller than standard ATX; usually fewer expansion slots.	Mainstream and compact desktops.
Mini-ITX	Compact board, commonly 170 × 170 mm.	Small-form-factor computers.

Compatibility Considerations

- CPU socket and supported processor generation must match.
- RAM type, supported capacity, speed and slot configuration must match the board.
- The case must support the motherboard form factor.
- The power supply must have suitable connectors and sufficient capacity.
- Storage interface and physical dimensions must be supported.
- Firmware version can affect support for newer processors or memory modules.

Advantages and Limitations

A motherboard makes a computer system integrated, expandable and easier to assemble. However, it does not make every component compatible with every other component. A failed motherboard can also affect many parts of the system, and replacing it may require checking CPU, RAM, case and power-supply compatibility.

Short Exam Answer: What is a motherboard?

The motherboard is the main printed circuit board of a computer. It connects the CPU, RAM, storage devices and peripheral interfaces, distributes power, and provides communication pathways so that the components can operate as one system.

2.2 Processor (CPU)

Definition and Role

The Central Processing Unit (CPU), commonly called the processor, executes program instructions. It performs calculations, makes logical comparisons, and coordinates operations with memory and input/output devices. It is often called the 'brain' of the computer, although the rest of the system is also essential.

Main Functions of the CPU

- Fetches instructions from memory.
- Decodes instructions to determine what action is required.
- Executes arithmetic, logical, data movement and control operations.
- Uses registers and cache to hold data and instructions temporarily.
- Coordinates with memory and input/output systems through control signals.
- Changes the flow of a program when an instruction requires a branch or decision.

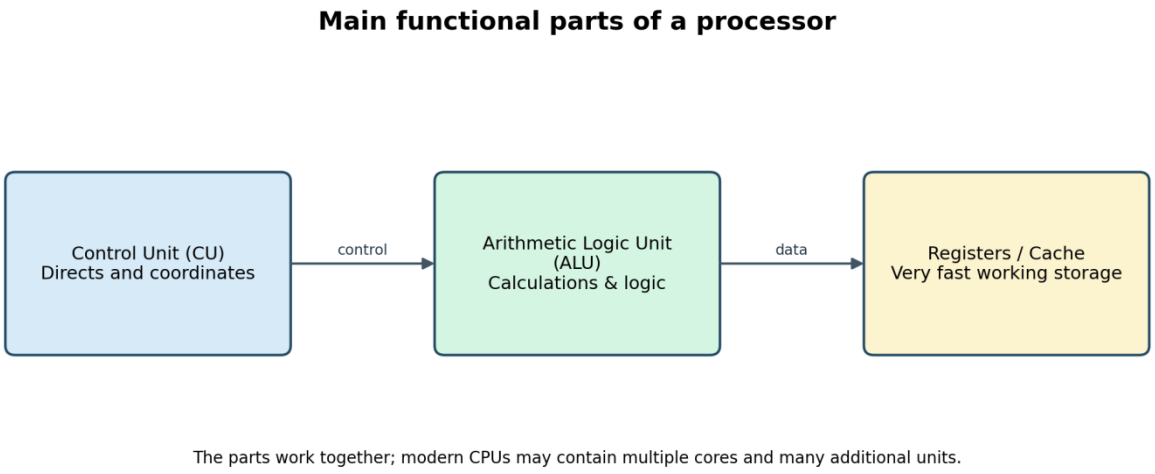


Figure 2. Simplified functional view of a CPU.

The Control Unit (CU)

The Control Unit directs and coordinates CPU operations. It interprets the decoded instruction and issues control signals that tell other parts of the processor and computer when to move data, perform an operation, or store a result.

- Controls the fetch-decode-execute process.
- Coordinates movement of instructions and data between registers, the ALU and memory interfaces.
- Produces timing and control signals required for operations.
- Ensures that instruction steps occur in the required order, including changes in program flow.

Remember for exams: The Control Unit controls and coordinates; it is not the part mainly responsible for arithmetic calculations.

The Arithmetic Logic Unit (ALU)

The Arithmetic Logic Unit performs arithmetic and logical operations on data. It is a core part of the CPU, although modern processors also contain specialised units for floating-point calculations, vector operations, cryptography and other tasks.

Operation type	Examples	Meaning
Arithmetic	Addition, subtraction, multiplication, division	Calculates numerical results.
Logical	AND, OR, NOT, XOR	Works with true/false conditions or individual bits.
Comparison	Equal to, greater than, less than	Helps a program make decisions.
Bit operations	Shift left, shift right, bit masking	Moves or tests individual bits.

Example: if a program asks whether 12 is greater than 7, the processor performs a comparison and sets status information used by the program's next step. The exact implementation depends on the instruction set and CPU design.

Control Unit vs ALU

Control Unit (CU)	Arithmetic Logic Unit (ALU)
Directs and coordinates operations.	Performs arithmetic and logical operations.
Interprets instructions and issues control signals.	Processes data values supplied to it.
Answers: what operation should happen, and when?	Answers: what is the result of this calculation or logic operation?

Machine Cycle (Instruction Cycle)

The machine cycle is the repeated sequence of steps used to process an instruction. Introductory textbooks commonly describe it as fetch, decode, execute and store/write back. Some books divide it into more stages, such as memory access or interrupt checking; the exact stages depend on the processor design.

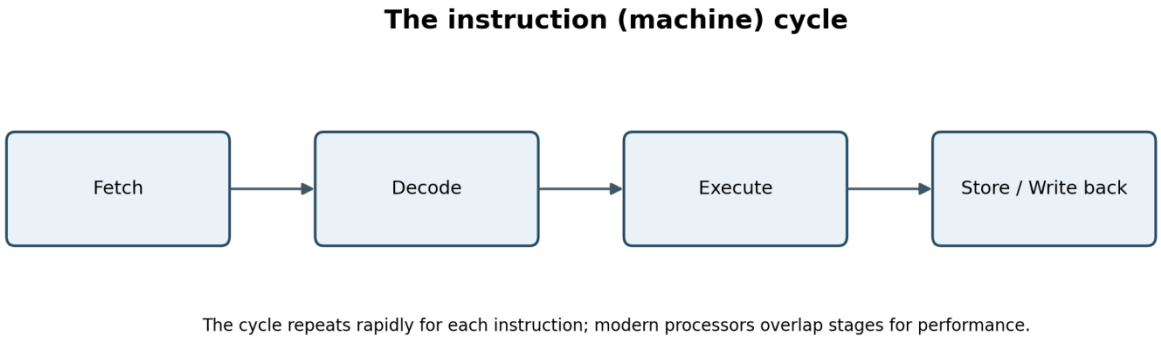


Figure 3. Basic instruction cycle.

7. Fetch: the processor obtains the next instruction from memory, using the address indicated by the program counter.
8. Decode: the control logic determines what the instruction means and what operands or resources it needs.
9. Execute: the processor carries out the required operation, such as adding values, comparing data or calculating an address.
10. Store / write back: the result is written to a register or memory when required, and the processor moves to the next instruction.

Example: for an instruction that adds 5 and 3, the CPU fetches the instruction, decodes it as an addition, executes the operation to produce 8, and writes the result to the required destination.

Exam wording: The machine cycle is not the same as one clock tick. An instruction may require multiple clock cycles, and modern CPUs may overlap the stages of several instructions using pipelining.

The System Clock

The system clock provides regular timing signals that help synchronise operations inside the computer. Clock frequency is measured in hertz (Hz). One gigahertz (GHz) means one billion cycles per second.

- A 3.0 GHz clock has a nominal clock frequency of three billion cycles per second.
- Clock frequency is not a direct measure of total performance. Two processors with the same frequency can perform differently.
- Performance also depends on architecture, instructions completed per cycle (IPC), core count, cache, memory, power limits, cooling and workload.
- Modern CPUs can change their clock speed dynamically to balance performance, temperature and power use.
- The CPU clock is not necessarily identical to every system bus or memory clock.

Cores, Threads and Cache

Term	Meaning	Why it matters
Core	A processing unit within a CPU capable of executing instructions.	More cores can improve workloads that can run in parallel.
Thread	A sequence of program instructions; CPU vendors also use the term for a logical processor exposed by simultaneous multithreading.	More logical threads can improve utilisation, but do not equal the same number of physical cores.
Cache	Small, fast memory close to or inside the CPU.	Reduces the need to fetch frequently used data from RAM.
IPC	Instructions per clock cycle, a measure of how much work a processor completes per cycle under particular conditions.	Helps explain why clock speed alone is insufficient.

Comparison of PC Processors

PC processors can be compared by manufacturer, product family, architecture, core/thread count, clock behaviour, cache, power consumption, integrated graphics, platform compatibility and real-world benchmark results. A product name or GHz value alone does not provide a complete comparison.

Comparison factor	What to check	Exam explanation
Manufacturer / family	Examples include Intel Core and AMD Ryzen families.	These are product families, not a guarantee that every model has equal performance.
Architecture / generation	The design generation and supported instruction features.	Newer designs may improve efficiency and performance, but gains vary.
Cores and threads	Physical cores and supported logical threads.	Useful for multitasking and parallel workloads.
Clock frequency	Base and boost frequencies, under stated power/thermal conditions.	Higher frequency can help, but is not a complete performance measure.
Cache	L1, L2 and L3 capacity and design.	Larger or better-designed cache can reduce memory delays.
Integrated graphics	Whether a graphics processor is built into the CPU/package.	Can avoid a separate graphics card for basic display and light graphics tasks.
Power and cooling	Thermal design/power specifications and actual system limits.	Affects heat, battery life, noise and sustained performance.

Socket and platform	Socket, chipset, firmware and memory support.	Determines whether the CPU can work with a particular motherboard.
Price and workload benchmarks	Results for the intended applications at a comparable price.	The best choice depends on the actual task, not the brand alone.

Intel and AMD: a fair comparison

Intel and AMD both make processors for desktop and laptop computers. Neither brand is always faster in every situation. Performance changes by model, generation, cooling, power limits and workload. Compare specific processors at similar prices and use independent benchmarks for tasks such as office work, gaming, compiling code or video editing.

Avoid this common mistake: Do not write 'Intel is always faster' or 'AMD is always cheaper'. These statements are too broad and can become incorrect as product generations change.

CPU Performance: Key Takeaway

CPU performance depends on several factors working together: architecture, cores, threads, clock behaviour, IPC, cache, memory performance, power and cooling. The right processor is the one that suits the workload and is compatible with the rest of the computer.

2.3 Memory

What is Computer Memory?

Computer memory stores data, instructions and results so that the processor can use them. In a broad sense, memory and storage technologies hold information in different ways. In basic computer studies, 'main memory' usually refers to RAM, while SSDs and hard drives are called secondary storage.

Memory vs Storage

Main memory (RAM)	Secondary storage (SSD / HDD)
Holds programs and data currently in use.	Keeps files and programs for long-term use.
Usually volatile: data is lost when power is removed.	Non-volatile: data remains without power.
Generally much faster than mass storage, but smaller in capacity.	Usually larger in capacity, with different speed and cost characteristics.
Example: a browser's active working data.	Example: saved documents, applications and operating system files.

Bits and Bytes

All digital information is represented using bits. A bit is the smallest basic unit of digital data and has one of two values: 0 or 1. A byte is a group of 8 bits. Bytes are commonly used to measure file sizes and memory capacity.

Unit	Meaning	Example
bit (b)	One binary digit: 0 or 1.	A single on/off-style value in a digital representation.
byte (B)	8 bits.	A small unit used to represent text or other data.
kilobyte (KB)	In decimal storage labels, 1 KB = 1,000 bytes.	Often used for small files.
megabyte (MB)	In decimal labels, 1 MB = 1,000 KB.	Photos, short audio files or small applications.
gigabyte (GB)	In decimal labels, 1 GB = 1,000 MB.	RAM capacity or large files.
terabyte (TB)	In decimal labels, 1 TB = 1,000 GB.	Large SSDs and hard drives.

Binary vs decimal units: Some textbooks use 1 KB = 1,024 bytes and then 1 MB = 1,024 KB. Strictly, these binary quantities are KiB, MiB and GiB. Read the convention used in your course or question, and state your assumption in calculations.

Quick Bit/Byte Conversions

- 1 byte = 8 bits.
- To convert bytes to bits, multiply by 8. Example: 5 bytes = 40 bits.
- To convert bits to bytes, divide by 8. Example: 64 bits = 8 bytes.
- Capitalisation matters: b usually means bit; B means byte. Network speeds are often given in Mbps, while file sizes are often given in MB.

Memory Addressing

Memory addressing is the method used to identify a particular location in memory. Each addressable location has an address, and the processor uses addresses to request data or instructions. In most modern general-purpose computers, memory is byte-addressable, meaning each distinct address identifies one byte.

An address is like a house number: it identifies where information is located, but it is not the information itself. The number of address bits determines how many distinct addresses can be represented.

Address bits	Number of unique addresses	If each address identifies 1 byte
8 bits	$2^8 = 256$	256 bytes of addressable space
16 bits	$2^{16} = 65,536$	64 KiB of addressable space
32 bits	$2^{32} = 4,294,967,296$	4 GiB of addressable space
64 bits	2^{64} possible address values in the simple theoretical model	The practical usable range depends on the CPU, operating system and implementation.

Important qualification: A 32-bit address space can represent 2^{32} byte addresses, which is 4 GiB in a byte-addressable model. Actual usable RAM can be lower because hardware regions and operating-system limits may reserve parts of the address space.

Types of Memory

1. RAM (Random Access Memory)

RAM is the computer's main working memory. It stores the programs, data and operating-system information currently being used. The CPU can access RAM locations directly, rather than reading every location in sequence.

- Volatile: contents are normally lost when power is removed.
- Read and write: programs and the operating system can update its contents.
- Faster than SSD/HDD storage but slower than CPU registers and cache.
- More RAM can help a computer keep more programs and data active without relying heavily on slower storage.
- Common modern system RAM is DRAM; desktop and laptop modules commonly use DDR-family memory.

Example: when a student opens a browser and a word processor, their active code and data are loaded into RAM. Saving a document writes it to persistent storage so it remains available after shutdown.

2. ROM (Read-Only Memory)

ROM is non-volatile memory used to retain information when power is off. Traditional ROM is written during manufacture and is not normally changed during ordinary use. The term is also used more broadly in introductory courses for firmware memory, although modern firmware is commonly stored in rewritable flash memory.

- Non-volatile: information remains without power.
- Traditionally used for fixed instructions or firmware.
- Different forms include mask ROM, PROM, EPROM and EEPROM.
- Modern PCs generally store UEFI firmware in flash memory, which can be updated.

3. Cache Memory

Cache is small, very fast memory located in or close to the CPU. It keeps copies of data and instructions that the processor is likely to use again. If the requested information is found in cache, the CPU can usually obtain it more quickly than by accessing RAM.

Cache level	General description
L1 cache	Usually the smallest and fastest cache, commonly dedicated to a core and split into instruction and data sections.
L2 cache	Usually larger than L1 and slightly slower; often associated with a core.
L3 cache	Often larger and shared by multiple cores on many CPUs; exact designs vary.

Cache works well because programs often show temporal locality (reusing recently accessed data) and spatial locality (accessing nearby data). A cache hit means the requested item is present; a cache miss means it must be fetched from a lower level of the memory hierarchy.

4. Flash Memory

Flash memory is non-volatile electronic memory that can be electrically erased and reprogrammed. It is widely used in USB flash drives, memory cards, SSDs, phones and embedded devices.

- Retains data without continuous power.
- Can be erased and rewritten, although cells have limited write/erase endurance.
- Often provides compact, shock-resistant storage with no moving mechanical parts.
- SSD performance depends on the controller, flash type, interface and workload.

Flash is a memory technology, not one single product. A USB drive and an SSD may both use flash memory but have different controllers, performance and reliability characteristics.

5. CMOS and CMOS/RTC Settings

CMOS stands for Complementary Metal-Oxide-Semiconductor, a technology used to build many electronic circuits. In traditional PC terminology, 'CMOS' is often used for the small battery-backed memory/settings associated with the real-time clock and firmware setup. Modern systems commonly store firmware settings in non-volatile memory, while the battery helps keep the real-time clock running and may support legacy settings behaviour.

- The real-time clock (RTC) keeps track of date and time when the computer is unplugged.
- A small coin-cell battery commonly supplies backup power to the RTC.
- If the battery is depleted, the computer may lose time or show a firmware settings warning.
- CMOS is not the same thing as RAM in general, and the term should not be used to mean all computer memory.

Exam-safe definition: CMOS, in traditional PC terminology, refers to the low-power, battery-backed circuitry/settings associated with the real-time clock and firmware configuration. The exact storage implementation varies on modern motherboards.

Memory Types at a Glance

Type	Volatile?	Main purpose	Can be changed?	Typical example
RAM	Yes	Active programs and data	Read/write during operation	DDR4/DDR5 system memory
ROM (traditional)	No	Fixed instructions/firmware	Not normally changed	Mask ROM
Flash	No	Persistent data and firmware	Electrically erased/reprogrammed	SSD, USB drive, firmware chip
Cache	Usually volatile	Speed up CPU access to reused data	Managed automatically by hardware	L1/L2/L3 cache
CMOS/RTC-backed settings	RTC needs backup power; modern settings may be stored non-volatilely	Clock and selected setup information	Through firmware setup or system mechanisms	Coin-cell battery and RTC circuit

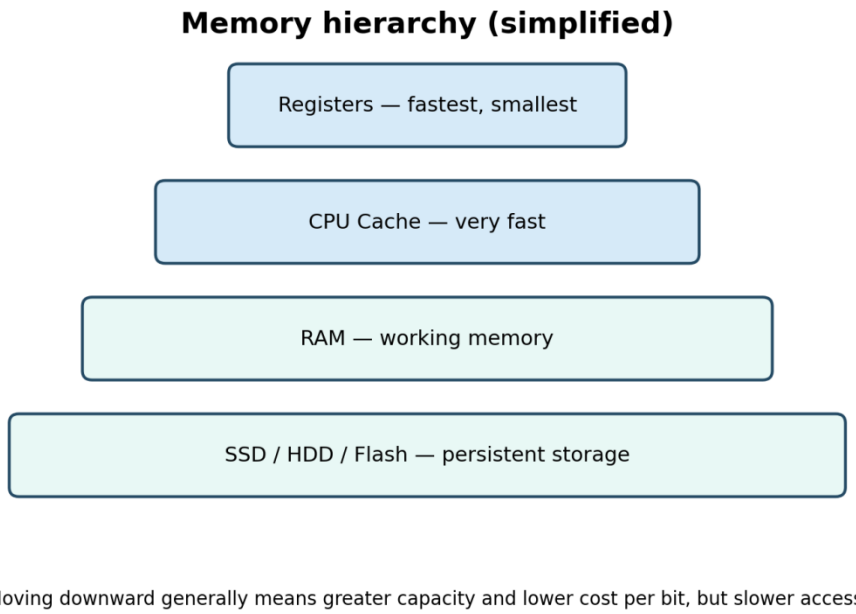


Figure 4. Simplified memory hierarchy. Persistent storage is shown for comparison with primary memory.

Volatile vs Non-Volatile Memory

Volatile memory	Non-volatile memory
Needs power to retain its contents.	Retains information without continuous power.
Examples: RAM, CPU registers and most CPU cache.	Examples: ROM, flash memory and SSD storage.
Commonly used for active work.	Commonly used for firmware and persistent storage.

How the Memory System Works Together

11. A program is stored on an SSD or hard drive.
12. When opened, the operating system loads the program and required data into RAM.
13. The CPU fetches instructions and data; frequently used items may be served from cache.
14. The CPU processes the instructions and keeps temporary values in registers and cache.
15. When the user saves a file, the operating system writes the updated data to persistent storage.

Quick Comparison Tables

Motherboard, CPU and Memory

Part	Primary role	Typical components / examples
Motherboard	Connects and enables communication between system components.	CPU socket, RAM slots, expansion slots, storage connectors.
CPU	Executes instructions and coordinates processing.	Control Unit, ALU, registers, cache, cores.
RAM	Holds active programs and data temporarily.	DDR-family memory modules.
Cache	Keeps frequently used data close to the CPU.	L1, L2 and L3.
Flash storage	Retains files without power.	SSD, USB flash drive, memory card.
Firmware memory	Stores startup firmware.	UEFI firmware commonly stored in flash.

Frequently Confused Terms

Do not confuse	Correct distinction
Motherboard vs CPU	The motherboard connects components; the CPU executes instructions.
CU vs ALU	CU directs operations; ALU performs arithmetic and logic.
Clock speed vs performance	Clock speed is frequency; overall performance also depends on architecture, IPC, cores, cache, power and workload.
RAM vs storage	RAM is working memory and usually volatile; SSD/HDD storage is persistent.
ROM vs flash	Traditional ROM is not normally rewritten; flash is non-volatile and can be electrically reprogrammed.
Bit vs byte	A bit is one binary digit; a byte is 8 bits.
Address vs data	An address identifies a location; data is the information stored or transferred.
Cache vs RAM	Cache is smaller and faster, usually managed by hardware; RAM is larger main memory.

Short Questions with Model Answers

1. Define a motherboard.

The motherboard is the main printed circuit board that connects the CPU, memory, storage and other components and provides pathways for communication.

2. Write any four functions of a motherboard.

It holds major components, provides data/control pathways, distributes power through board circuitry and connectors, and provides interfaces for storage and peripherals.

3. What is a CPU?

The CPU is the processor that fetches, decodes and executes program instructions and performs processing and control operations.

4. What is the function of the Control Unit?

The Control Unit interprets instructions and coordinates CPU operations by issuing control signals.

5. What is the ALU?

The Arithmetic Logic Unit performs arithmetic calculations, logical operations and comparisons.

6. List the basic stages of the machine cycle.

Fetch, decode, execute and store/write back.

7. Define system clock.

The system clock produces regular timing signals used to synchronise operations. Its frequency is measured in hertz.

8. Why is GHz alone not enough to compare CPUs?

Because performance also depends on architecture, IPC, cores, cache, power limits, cooling and workload.

9. What is a bit?

A bit is a binary digit that can represent 0 or 1.

10. How many bits are in one byte?

Eight bits.

11. What is memory addressing?

It is the method of identifying memory locations by assigning each addressable location a unique address.

12. Define RAM.

RAM is volatile read/write main memory used to hold programs and data currently in use.

13. Define ROM.

ROM is non-volatile memory traditionally used to store fixed instructions or firmware.

14. What is cache memory?

Cache is small, fast memory close to the CPU that stores copies of frequently or recently used data and instructions.

15. What is flash memory?

Flash is non-volatile memory that can be electrically erased and reprogrammed, used in SSDs, USB drives and firmware storage.

16. What is CMOS in a PC?

The term traditionally refers to low-power, battery-backed circuitry/settings associated with the RTC and firmware configuration.

17. Differentiate volatile and non-volatile memory.

Volatile memory loses its contents when power is removed; non-volatile memory retains data without power.

18. What is the purpose of a CPU socket?

It physically and electrically connects a compatible processor to the motherboard.

19. What is a cache hit?

A cache hit occurs when the requested data or instruction is found in the cache.

20. What does a 32-bit byte address space represent?

In the simple byte-addressable model, 2^{32} distinct addresses, equal to 4 GiB of addressable space.

Long Questions: Suggested Answer Plans

1. Explain the motherboard and its main components.

Suggested structure: Definition → labelled diagram → CPU socket → RAM slots → chipset/I/O → expansion slots → storage connectors → power connectors → firmware chip → ports → compatibility and conclusion.

2. Explain the CPU, Control Unit and ALU.

Suggested structure: Define CPU → state CPU functions → explain CU and its control role → explain ALU with arithmetic/logical examples → compare CU and ALU → add CPU block diagram.

3. Describe the machine cycle.

Suggested structure: Define instruction cycle → draw Fetch → Decode → Execute → Store/Write back → explain each stage → give a simple addition example → note that exact stages vary by architecture.

4. Explain the system clock and factors affecting CPU performance.

Suggested structure: Define clock and Hz/GHz → explain synchronisation → explain why clock speed is not equal to performance → discuss architecture, IPC, cores, threads, cache, power and cooling.

5. Explain bits, bytes and memory addressing.

Suggested structure: Define bit and byte → give 1 byte = 8 bits → explain decimal/binary units → define address → explain address bits and 2^n addresses → solve a simple 32-bit address-space example.

6. Compare RAM, ROM, cache, flash and CMOS.

Suggested structure: Define memory → use a comparison table → state volatility, purpose, speed/usage and examples → clarify that modern firmware commonly uses flash and CMOS is often a legacy term for RTC/settings.

Practice MCQs

1. Which component is the main circuit board of a computer?

A. ALU B. Motherboard C. RAM D. Cache

2. Which CPU part performs arithmetic and logical operations?

A. CU B. PSU C. ALU D. ROM

3. Which sequence best represents the basic instruction cycle?

A. Store, fetch, print, decode B. Fetch, decode, execute, store C. Decode, save, fetch, delete D. Input, output, shutdown, boot

4. CPU clock frequency is measured in:

A. Bytes B. Volts only C. Hertz D. Pixels

5. Which memory is normally volatile?

A. RAM B. Flash drive C. SSD D. ROM

6. One byte contains:

A. 2 bits B. 4 bits C. 8 bits D. 16 bits

7. Which memory is designed to reduce average CPU access time?

A. Cache B. USB storage C. Optical disc D. Printer memory only

8. Which is a common use of flash memory?

A. Mechanical fan rotation B. SSD storage C. CPU heatsink D. Monitor stand

9. In a byte-addressable system, an address identifies:

A. A colour B. A byte location C. A CPU brand D. A clock speed

10. Which statement about CPU comparison is most accurate?

A. GHz alone determines speed B. One brand is always fastest C. Architecture and workload also matter D. More threads always mean twice the performance

11. The Control Unit mainly:

A. Coordinates operations B. Stores files permanently C. Displays images D. Supplies wall power

12. Which statement about traditional ROM is correct?

A. It is always volatile B. It is traditionally not changed during normal use C. It is the same as cache D. It is always a mechanical drive

MCQ Answer Key

1-B, 2-C, 3-B, 4-C, 5-A, 6-C, 7-A, 8-B, 9-B, 10-C, 11-A, 12-B.

Final One-Page Revision Sheet

- Motherboard = main circuit board; connects components and supports communication.
- CPU = executes instructions. CU directs; ALU calculates and compares.
- Machine cycle = fetch → decode → execute → store/write back.
- System clock = timing signal; frequency in Hz/GHz. Clock speed alone does not determine performance.
- Core = physical processing unit; thread = instruction sequence/logical execution context; cache = small, fast memory close to CPU.
- 1 byte = 8 bits. b usually means bit; B means byte.
- Address identifies a location; data is the content. With n address bits, there are 2^n possible address values.
- RAM = volatile working memory; ROM = traditionally fixed non-volatile memory; flash = rewritable non-volatile memory.
- Cache improves average access speed by keeping likely-to-be-reused information close to the CPU.
- CMOS is traditionally associated with battery-backed RTC/settings; modern firmware settings may be stored in non-volatile memory.
- When comparing processors, consider architecture, cores/threads, IPC, cache, clock behaviour, power, cooling, price and the target workload.

Glossary

Term	Simple meaning
ALU	Part of the CPU that performs arithmetic and logic.
Address	Identifier used to locate a memory position or device.
Bus	Pathway or interconnect that carries data, addresses or control signals.
Cache hit/miss	Requested data is found/not found in a cache level.
Clock frequency	Number of clock cycles per second.
Control Unit	CPU section that directs and coordinates operations.

Firmware	Low-level software used to initialise or control hardware.
Memory hierarchy	Arrangement of storage levels with different speed, size and cost.
Non-volatile	Retains information when power is off.
PCB	Printed circuit board containing electronic components and connections.
Volatile	Normally loses stored contents when power is removed.

End of Notes